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Method and apparatus to improve performance of a redundant array of independent disks that includes zoned namespaces drives

Abstract

High performance parity-based Redundant Array of Independent Disks (RAID) on Zoned Namespaces Solid State Drives (SSD)s with support for high queue depth write Input Output (IO) and Zone Append command is provided in a host system. The host system includes a stripe mapping table to store mappings between parity strips and data strips in stripes on the RAID member SSDs. The host system also includes a Logical to Physical (L2P) table to store data block addresses returned by the Zone Append command.

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Background/Summary

FIELD

(1) This disclosure relates to redundant array of independent disks (RAID) systems and in particular to a RAID system with zoned namespaces drives.

BACKGROUND

(2) A Redundant Array of Independent Disks (RAID) combines a plurality of physical drives (hard disk drives (HDDs) or solid state drives (SSDs)) into a logical drive for purposes of reliability, capacity, or performance. Instead of multiple physical drives, an operating system sees the single logical drive. As is well known to those skilled in the art, there are many standard methods referred to as RAID levels for distributing data across the physical drives in a RAID system.

(3) For example, in a level 0 RAID system the data is striped across a physical array of drives by breaking the data into blocks and writing each block to a separate drive. Input/Output (I/O) performance is improved by spreading the load across many drives. Although a level 0 RAID improves I/O performance, it does not provide redundancy because if one drive fails, all of the data is lost

(4) A level 5 RAID system provides a high level of redundancy by striping both data and parity information across at least three drives. Data striping is combined with distributed parity to provide a recovery path in case of failure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features of embodiments of the claimed subject matter will become apparent as the following detailed description proceeds, and upon reference to the drawings, in which like numerals depict like parts, and in which:

(2) FIG. 1 is a block diagram illustrating the layout of stripes including data strips and parity strips in a level 5 RAID system with three member drives;

(3) FIG. 2 is a block diagram of a storage system that includes the level 5 RAID system shown in FIG. 1 using SSDs with Zoned Namespaces;

(4) FIG. 3 is a block diagram illustrating stripe mapping entries in the stripe mapping table to map data strips and parity strips for stripes to RAID member drives;

(5) FIG. 4 is a block diagram illustrating the use of the stripe mapping entries in the stripe mapping table to perform a level 5 RAID rebuild after failure of a level 5 RAID member drive;

(6) FIG. 5 is a flowgraph illustrating a write operation in the level 5 RAID system that uses SSDs with Zoned Namespaces;

(7) FIG. 6 is a flowgraph illustrating a RAID rebuild the level 5 RAID system that uses SSDs with Zoned Namespaces; and

(8) FIG. 7 is a block diagram of an embodiment of a computer system that includes a RAID system and a Stripe Mapping Table.

(9) Although the following Detailed Description will proceed with reference being made to illustrative embodiments of the claimed subject matter, many alternatives, modifications, and variations thereof will be apparent to those skilled in the art. Accordingly, it is intended that the claimed subject matter be viewed broadly, and be defined as set forth in the accompanying claims.

DESCRIPTION OF EMBODIMENTS

(10) FIG. 1 is a block diagram illustrating the layout of stripes **104-1**, **104-2**, **104-3** including data strips **106** and parity strips **108** in a level 5 RAID system **100** with three member drives **102-1**, **102-2**, **102-3**.

(11) The level 5 RAID system **100** shown in FIG. 1 has three drives **102-1**, **102-2**, **102-3**. Three stripes **104-1**, **104-2**, **104-3** are shown, with each stripe **104-1**, **104-2**, **104-3** including two data strips **106** and one parity strip **108**.

- (12) Data is written to the level 5 RAID system **100** using block-level striping with parity distributed across the disks in a round robin fashion. Sequential data, for example, a file segmented into blocks may be distributed across a stripe, for example, horizontal stripe **104-1**, with data block **D1** stored in data strip **106** on drive **102-1**, data block **D2** stored in data strip **106** on drive **102-2** and parity **P12** computed for data block **D1** and data block **D2** stored in parity **P12** in parity strip **108** on drive **102-3** in stripe **104-1**.
- (13) Data block **D3** is stored in data strip **106** on drive **102-1**, data block **D4** is stored in data strip **106** on drive **102-3** and parity **P34** computed for data block **D3** and data block **D4** stored in parity **P34** in parity strip **108** on drive **102-2** in stripe **104-2**.
- (14) Data block **D5** is stored in data strip **106** on drive **102-2**, data block **D6** is stored in data strip **106** on drive **102-3** and parity **P56** computed for data block **D5** and data block **D6** is stored in parity **P56** in parity strip **108** on drive **102-1** in stripe **104-3**.
- (15) To write data to stripes **104-1**, **104-2**, **104-3**, the parity (**P12**) for data block **D1** and data block **D2**, parity (**P34**) for data block **D3** and data block **D4** and parity (**P56**) for data block **D5** and data block **D6** is computed in a host system prior to sending write commands to the level 5 RAID system **100**. The Logical Block addresses for the parity strips (**P12**, **P34**, **P56**) and the data strips (**D1**, **D2**, **D3**, **D4**, **D5**, **D6**) are selected by a driver in an operating system in the host system. An operating system is software that manages computer hardware and software including memory allocation and access to I/O devices. Examples of operating systems include Microsoft® Windows®, Linux®, iOS® and Android®.
- (16) After the parity has been computed and the logical block addresses selected, the host system sends write commands for the data strips and parity strips with the selected logical block addresses to each of the drives **102-1**, **102-2**, **102-3**. The write commands can be sent with a write queue depth greater than one to achieve high write performance.
- (17) The parity strips (**P12**, **P34**, **P56**) and the data strips (**D1**, **D2**, **D3**, **D4**, **D5**, **D6**) are stored in stripes **104-1**, **104-2**, **104-3** on the three member drives **102-1**, **102-2**, **102-3** as shown in the FIG. 1, so that the parity strip in the stripe is the parity for the data strips in the stripe. If a single member drive fails, an EXclusive OR (XOR) operation can be performed on the strips (two data strips or one data strip and a parity strip) for the stripe stored in the non-failed drives to recover the strip (data or parity) on the failed drive.
- (18) Non-Volatile Memory Express (NVMe®) specifications define a register level interface for host software to communicate with a non-volatile memory subsystem (for example, a Solid-state Drive (SSD)) over Peripheral Component Interconnect Express (PCIe), a high-speed serial computer expansion bus). NVMe is the industry standard for SSDs. The NVM Express specifications are available at www.nvmexpress.org. The PCIe standards are available at www.pcisig.com.
- (19) The NVM Express® (NVMe®) Base specification defines an interface for host software to communicate with a non-volatile memory subsystem over a variety of memory based transports and message based transports. The NVM Express® Zoned Namespace Command Set Specification defines a specific NVMe I/O Command Set, the Zoned Namespace Command Set, which extends the NVMe Base Specification and the NVM Command Set Specification.
- (20) A zoned namespace is a collection of non-volatile memory that is divided into a set of equally-sized zones. The zones are contiguous non-overlapping ranges of logical block addresses. Each zone has an associated Zone Descriptor that contains a set of attributes.
- (21) Zoned Namespace SSDs are highly optimized for capacity. However, Zoned Namespace SSDs expose a sequential-only interface for write Input/Output with a queue-depth of 1. The Zoned Namespace Command Set includes a Zone Append command that allows write I/O for a queue-depth greater than one.
- (22) The level 5 RAID system **100** relies on the placement of data strips **106** and parity strips **108** in stripes **104-1**, **104-2**, **104-3** on the RAID member drives **102-1**, **102-2**, **102-3** to locate data and

parity when data is read from the level 5 RAID system **100** and for a RAID rebuild process that needs to match data strips with parity strips when there is a drive failure.

(23) However, the Zone Append command does not allow the Logical Block Addresses for the parity strips (**P12**, **P34**, **P56**) and the data strips (**D1**, **D2**, **D3**, **D4**, **D5**, **D6**) to be selected by a driver in the host system. The Logical Block Addresses are returned by the Zone Namespace SSD after the data has been written to the Zoned Namespace SSD. Using the Zone Append command with queue depth greater than one, the writing of the data strips and parity strips can be reordered and different logical block addresses can be returned from each SSD for the parity strips and data strips for the same stripe. If a single member drive fails, an eXclusive OR (XOR) operation cannot be performed on the strips (two data strips or one data strip and a parity strip) for the stripe because some of the strips no longer correspond from the Logical Block Address perspective.

(24) To ensure that the strips (parity and data) for the same stripe are not reordered in the Zone Namespace SSDs, the write sequence is performed using write I/O with queue-depth **1**. Referring to FIG. **1**, the parity (**P12**) for data block **D1** and data block **D2**, parity (**P34**) for data block **D3** and data block **D4** and parity (**P56**) for data block **D5** and data block **D6** is computed in a host system prior to sending write commands to the level 5 RAID system **100**.

(25) A first write command for data strip **D1** is sent to member drive **102-1**, a first write command for data strip **D2** is sent to member drive **102-2**, and a first write command for parity strip **P12** is sent to member drive **102-3**. An LBA is returned from each member drive after the data/parity strip has been written.

(26) After the first write commands have been completed by each of the member drives, a second write command for data strip **D3** is sent to member drive **102-1**, a second write command for data strip **D4** is sent to member drive **102-3**, and a second write command for parity strip **P34** is sent to member drive **102-2**. A LBA is returned from each member drive after the data/parity strip has been written.

(27) After the second write commands have been completed by each of the member drives, a third write command for data strip **D5** is sent to member drive **102-2**, a third write command for data strip **D6** is sent to member drive **102-2**, and a third write command for parity strip **P56** is sent to member drive **102-1**. A LBA is returned from each member drive after the data/parity strip has been written.

(28) Performing the write sequence using write I/O with queue-depth **1** allows data to be recovered for a failed member of the RAID. However, performance of a RAID with queue-depth **1** is low.

(29) High performance parity-based RAID on Zoned Namespaces SSDs with support for high queue depth write IO and Zone Append command is provided in a host system. The host system includes a stripe mapping table to store mappings between parity strips and data strips in stripes on the RAID member SSDs. The host system also includes a Logical to Physical (L2P) table to store data block addresses returned by the Zone Append command.

(30) Various embodiments and aspects of the inventions will be described with reference to details discussed below, and the accompanying drawings will illustrate the various embodiments. The following description and drawings are illustrative of the invention and are not to be construed as limiting the invention. Numerous specific details are described to provide a thorough understanding of various embodiments of the present invention. However, in certain instances, well-known or conventional details are not described in order to provide a concise discussion of embodiments of the present inventions.

(31) Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in conjunction with the embodiment can be included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification do not necessarily all refer to the same embodiment.

(32) FIG. **2** is a block diagram of a storage system **200** that includes the level 5 RAID system **100**

shown in FIG. 1 using SSDs with Zoned Namespaces. The storage system **200** includes a RAID controller **212** and a Random Access Media **210** to store a Logical-to-Physical (L2P) mapping table **206**, a write buffer **208** and a stripe mapping table **204**.

(33) A non-volatile memory (NVM) device is a memory whose state is determinate even if power is interrupted to the device. The block addressable non-volatile memory can be a NAND Flash memory, or more specifically, multi-threshold level NAND flash memory (for example, Single-Level Cell (“SLC”), Multi-Level Cell (“MLC”), Tri-Level Cell (“TLC”), Quad-Level Cell (“QLC”), Penta-Level Cell (“PLC”) or some other NAND Flash memory).

(34) A logical block is the smallest addressable data unit for read and write commands to access block addressable non-volatile memory in the solid state drives **102-1**, **102-2**, **102-3**. The address of the logical block is commonly referred to as a Logical Block Address (LBA). The L2P mapping table **206** can also be referred to as an L2P address indirection table, an L2P address table or an L2P table. The L2P mapping table **206** stores a physical block address in block addressable non-volatile memory in the zoned namespace solid state drives **102-1**, **102-2**, **102-3** corresponding to each LBA. The random access media **210** can be a byte-addressable persistent memory, for example, 3D) XPoint (Intel® Optane® Persistent Memory).

(35) Data from application **202** to be written to the level 5 RAID system **100** is stored in the write buffer **208** in the random access media **210**. A stripe mapping table **204** in the non-volatile random access media **210** stores the structure of stripes storing data written to the level 5 RAID system **100**. The stripe mapping table **204** is updated with the relevant mapping of strips and stripes in the member solid state drives **102-1**, **102-2**, **102-3** after the data strips and parity strips have been written to the RAID member drives **102-1**, **102-2**, **102-3** and each member drive returns the assigned LBAs for the data strips and the data stripes.

(36) In the example shown in FIG. 2, the Zone Append command has been used to write data to the RAID member drives **102-1**, **102-2**, **102-3** resulting in data strips and parity strips being misaligned in stripes **S1**, **S2**, **S3**, so that the RAID stripes are not stored in a standard RAID 5 layout as shown in the example in FIG. 1.

(37) The Zone Append command writes data and metadata, if applicable, to the Input/Output (I/O) controller in the Solid State Drive for the zone indicated by a Zone Select Logical Block Address (ZSLBA) field included with the Zone Append command. The I/O controller assigns the data and metadata, if applicable, to a set of logical blocks within the zone. The lowest LBA of the set of logical blocks written is returned in a completion queue entry.

(38) Referring to FIG. 2, each stripe includes two data strips and a parity strip. A first stripe includes data strip **D1 250** in RAID member drive **1 102-1**, data strip **D2 252** in RAID member drive **2 102-2** and parity strip **P12 260** in RAID member drive **102-3**. A second stripe includes data strip **D3 262** in RAID member drive **1 102-1**, data strip **D4 254** in RAID member drive **3 102-3** and parity strip **P34 258** in RAID member drive **102-2**. A third stripe includes data strip **D5 264** in RAID member drive **2 102-2**, data strip **D6 266** in RAID member drive **3 102-3** and parity strip **D56 256** in RAID member drive **102-1**. The data block locations (logical address) of data strips and parity strips in member drives **102-1**, **102-2**, **102-3** that are returned in the completion queue entries for the respective Zone Append Commands sent to each of the member drives **102-1**, **102-2**, **102-3** are stored in the L2P mapping table **206**. An entry in the L2P mapping table **206** stores the logical address of a data strip or parity strip in a member drive. The logical address of the data strip or parity strip is mapped to a physical address in the member drive by the member drive.

(39) FIG. 3 is a block diagram illustrating stripe mapping entries **302**, **304**, **306** in the stripe mapping table **204** to map data strips and parity strips for stripes to RAID member drives **102-1**, **102-2**, **102-3**.

(40) The stripe mapping table **204** is a contiguous memory space to store numbers that represent the mapping between a stripe in the RAID system and the strips in the members of the RAID system that correspond to the stripe in the RAID system. One number is stored for every strip in the

RAID system. The numbers stored in the stripe mapping table **204** represent the strip number on the member of the RAID system. In addition to storing the numbers, the stripe mapping table **204** can also include an array of pointers **300**, with each pointer in the array of pointers **300** to store the location of the number of the first strip in each stripe stored in the stripe mapping table **204**.

(41) In the example shown in FIG. 3, the stripe mapping entry **302** for stripe **1** stores three numbers representing the strips for stripe **1** stored on RAID member drives **102-1**, **102-2**, **102-3**. The first strip **308** for stripe **1** is stored in strip number **1** (D1 250) on drive **1 102-1**, the second strip **310** for stripe **1** is stored in strip number **1** (D2 252) on drive **2 102-2** and the third strip **312** for stripe **1 302** is stored in strip number **2** (P12 260) on drive **3 102-3**.

(42) The stripe mapping entry **304** for stripe **2** stores three numbers representing the strips for stripe **2** stored on RAID member drives **102-1**, **102-2**, **102-3**. The first strip **314** for stripe **2** is stored in strip number **3** (D3 262) on drive **1 102-1**, the second strip **316** for stripe **2** is stored in strip number **2** (P34 258) on drive **2 102-2** and the third strip **318** for stripe **2** is stored in strip number **1** (D4 260) on drive **3 102-3**.

(43) The stripe mapping entry **306** for stripe **3** stores three numbers representing the strips for stripe **3** stored on RAID member drives **102-1**, **102-2**, **102-3**. The first strip **320** for stripe **3** is stored in strip number **2** (P56 256) on drive **1 102-1**, the second strip **322** for stripe **3** is stored in strip number **3** (D5 264) on drive **2 102-2** and the third strip **324** for stripe **3** is stored in strip number **3** (D6 266) on drive **3 102-3**.

(44) FIG. 4 is a block diagram illustrating the use of the stripe mapping entries **302**, **304**, **306** in the stripe mapping table **204** to perform a RAID rebuild after failure of a RAID member drive.

(45) In the example shown in FIG. 4, RAID member drive **3 102-3** has failed. A RAID rebuild is performed on a replacement drive for RAID member drive **3 102-3**. Strips that are stored on the failed RAID member drive **3 102-3** are recovered from the strips that are stored on non-failed RAID member drive **1 102-1** and RAID member drive **2 102-2**. The recovered strips are written to the replacement drive sequentially, starting with strip number **1**.

(46) The array of stripe pointers **300** is modified, so the strip numbers stored on the failed RAID member drive **3 102-3** are sorted in ascending order. This allows the strips stored on the failed RAID member drive **3 102-3** to be recovered in the correct order and written to the replacement drive sequentially, starting with strip number **1**.

(47) As shown in the example in FIG. 4, stripe pointer **S1** is modified to store the location of the number of the first strip in the stripe that stores strip **1** (D4 254) in failed RAID member drive **102-3** and stripe pointer **S2** is modified to store the location of the number of the first strip in the stripe that stores strip **2** (P12 260) in failed member drive **102-3**. Stripe pointer **S3** stores the location of the number of the first strip in the stripe that stores strip **3** (D6 266) in failed member drive **102-3**. Thus, stripes are processed such that strips stored in failed RAID member drive **102-3** are recovered sequentially and written sequentially to the replacement drive.

(48) In an embodiment in which the stripe mapping table **204** does not include an array of pointers, the stripe mapping table **204** is searched and sorted on the fly during rebuild of the failed member drive **102-3** on the replacement drive instead of prior to the rebuild. A per stripe search is performed in the stripe mapping table **204** in the RAID system to recover each strip prior to storing each recovered strip on the replacement drive.

(49) FIG. 5 is a flowgraph illustrating a write operation in the level 5 RAID system that uses SSDs with Zoned Namespaces.

(50) At block **500**, if a write request to write data to the level 5 RAID system, processing continues with block **502**. If not, processing continues with block **500**.

(51) At block **502**, data to be written to the level 5 RAID system is stored in the write buffer **208**.

(52) At block **504**, the L2P mapping table **206** is updated with the LBAs of the data stripes stored in the write buffer **208** to be written to the RAID volume.

(53) At block **506**, prior to writing the data stored in the write buffer **208** to the level 5 RAID

system, the RAID controller **212** generates the parity strip for the two data strips to be stored in a stripe on the member drives **102-1, 102-2, 102-3**.

(54) At block **508**, after the parity strip has been generated, the RAID controller **212** sends a Zone Append command to the member drives **102-1, 102-2, 102-3** to write the parity strip and data strips to the member drives **102-1, 102-2, 102-3**.

(55) At block **510**, after each of the member drives **102-1, 102-2, 102-3** has completed the write operations, each member drive returns the LISA assigned to the data strip or parity strip in the respective member drive to the RAID controller **212**.

(56) At block **512**, the RAID controller **212** updates the LBAs in the mapping table **206** with the LBAs on the RAID volume that are assigned to the data strips for each stripe by the RAID member drives **102-1, 102-2, 102-3**.

(57) At block **514**, the Stripe Mapping Table **204** is updated with the number of each strip in each member drive **102-1, 102-2, 102-3** associated with each stripe written to the RAID member drives as discussed in the example shown in FIG. 3.

(58) At block **516**, the Stripe Mapping Table **204** can be written to persistent memory that can be in a solid state drive (persistent storage),

(59) FIG. 6 is a flowgraph illustrating a RAID rebuild in the level 5 RAID system that uses SSDs with Zoned Namespaces.

(60) At block **600**, upon detection that one of the members of the RAID system has failed, a RAID rebuild operation is initiated. The failed drive is replaced by a replacement drive. If the Stripe Mapping Table is stored in persistent memory or a storage device and is not in Dynamic Random Access Memory, it is copied from the persistent memory or the storage device to the Dynamic Random Access Memory in Random Access Media **210**.

(61) At block **602**, the array of stripe pointers **300** is modified, so that the strip numbers stored on the failed RAID member drive are sorted in ascending order.

(62) At block **604**, if there is a SMT entry for a strip in the replacement drive that has not been written to the replacement drive, processing continues with block **606**. If all strips stored on the failed drive have been written to the replacement drive, processing continues with block **612**.

(63) At block **606**, the next SMT entry for a stripe is read from the Stripe Mapping Table **204**.

(64) At block **608**, the strips for the stripe that are stored on the non-failed drives are read from the non-failed drives.

(65) At block **610**, an XOR operation is performed on the data stored in the two strips read from the non-failed drives to generate the data stored in the strip in failed drive. The result of the XOR operation is stored in the strip in the replacement drive.

(66) At block **612**, the rebuild of the replacement drive is complete.

(67) An example has been described for a RAID level 5 system. Other parity-based RAID levels, for example, a level 6 RAID system can also use Zone Namespace Solid State Drives and a Stripe Mapping Table. A level 6 RAID system includes two parity drives (P and Q) and can recover from a failure of two drives. The Stripe Mapping Table includes two sets of pointers, one set for each failed drive to recover from the failure of two drives at the same time. Each set of pointers is sorted in ascending order for one of the failed drives so that each of the failed drives can be rebuilt in a sequential manner.

(68) High queue-depth workloads are enabled for RAID system using Zoned Namespace solid state drives, within a single zone allowing the full performance of Zoned Namespace drives for a RAID system. The memory used to store the SMT table is minimal because only one number is stored per RAID strip. For example, 4 Giga Bytes (GB) is used to store the SMT table with one 64-bit number stored per RAID strip for a level RAID system with three 16 Tera Bytes (TB) Solid State Drives with each strip to store 128 Kilo Bytes (KB).

(69) An example of a RAID system with solid state drives has been described. In other embodiments, the RAID system can include hard disk drives.

(70) FIG. 7 is a block diagram of an embodiment of a computer system **700** that includes a RAID system **100** and a Stripe Mapping Table **204**. Computer system **700** can correspond to a computing device including, but not limited to, a server, a workstation computer, a desktop computer, a laptop computer, and/or a tablet computer.

(71) The computer system **700** includes a system on chip (SOC or SoC) **704** which combines processor, graphics, memory, and Input/Output (I/O) control logic into one SoC package. The SoC **704** includes at least one Central Processing Unit (CPU) module **708**, a volatile memory controller **714**, and a Graphics Processor Unit (GPU) **710**. In other embodiments, the volatile memory controller **714** can be external to the SoC **704**. The CPU module **708** includes at least one processor core **702** and a level 2 (L2) cache **706**.

(72) Although not shown, each of the processor core(s) **702** can internally include one or more instruction/data caches, execution units, prefetch buffers, instruction queues, branch address calculation units, instruction decoders, floating point units, retirement units, etc. The CPU module **708** can correspond to a single core or a multi-core general purpose processor, such as those provided by Intel® Corporation, according to one embodiment.

(73) The Graphics Processor Unit (GPU) **710** can include one or more GPU cores and a GPU cache which can store graphics related data for the GPU core. The GPU core can internally include one or more execution units and one or more instruction and data caches. Additionally, the Graphics Processor Unit (GPU) **710** can contain other graphics logic units that are not shown in FIG. 7, such as one or more vertex processing units, rasterization units, media processing units, and codecs.

(74) Within the I/O subsystem **712**, one or more I/O adapter(s) **716** are present to translate a host communication protocol utilized within the processor core(s) **702** to a protocol compatible with particular I/O devices. Some of the protocols that adapters can be utilized for translation include Peripheral Component Interconnect (PCI)-Express (PCIe); Universal Serial Bus (USB); Serial Advanced Technology Attachment (SATA) and Institute of Electrical and Electronics Engineers (IEEE) 1594 “Firewire”.

(75) The I/O adapter(s) **716** can communicate with external I/O devices **724** which can include, for example, user interface device(s) including a display and/or a touch-screen display **752**, printer, keypad, keyboard, communication logic, wired and/or wireless, storage device(s) including hard disk drives (“HDD”), solid-state drives (“SSD”), removable storage media, Digital Video Disk (DVD) drive, Compact Disk (CD) drive, Redundant Array of Independent Disks (RAID), tape drive or other storage device. The storage devices can be communicatively and/or physically coupled together through one or more buses using one or more of a variety of protocols including, but not limited to, SAS (Serial Attached SCSI (Small Computer System Interface)), PCIe (Peripheral Component Interconnect Express), NVMe (NVM Express) over PCIe (Peripheral Component Interconnect Express), and SATA (Serial ATA (Advanced Technology Attachment)).

(76) Additionally, there can be one or more wireless protocol I/O adapters. Examples of wireless protocols, among others, are used in personal area networks, such as IEEE 802.15 and Bluetooth, 4.0; wireless local area networks, such as IEEE 802.11-based wireless protocols; and cellular protocols.

(77) The I/O adapter(s) **716** can also communicate with a level 5 RAID system **100** with three member drives **102-1**, **102-2**, **102-3**. The level 5 RAID system can include a Peripheral Component Interconnect Express (PCIe) adapter that is communicatively coupled using the NVMe (NVM Express) over PCIe (Peripheral Component Interconnect Express) protocol over a bus to the member drives **102-1**, **102-2**, **102-3**. Non-Volatile Memory Express (NVMe) standards define a register level interface for host software to communicate with a non-volatile memory subsystem (for example, a Solid-state Drive (SSD)) over Peripheral Component Interconnect Express (PCIe), a high-speed serial computer expansion bus). The NVM Express standards are available at www.nvmexpress.org. The PCIe standards are available at www.pcisig.com.

(78) NVM device **750** can include a byte-addressable write-in-place three dimensional crosspoint

memory device, or other byte addressable write-in-place NVM devices (also referred to as persistent memory), such as single or multi-level Phase Change Memory (PCM) or phase change memory with a switch (PCMS), NVM devices that use chalcogenide phase change material (for example, chalcogenide glass), resistive memory including metal oxide base, oxygen vacancy base and Conductive Bridge Random Access Memory (CB-RAM), nanowire memory, ferroelectric random access memory (FeRAM, FRAM), magneto resistive random access memory (MRAM) that incorporates memristor technology, spin transfer torque (STT)-MRAM, a spintronic magnetic junction memory based device, a magnetic tunneling junction (MTJ) based device, a DW (Domain Wall) and SOT (Spin Orbit Transfer) based device, a thyristor based memory device, or a combination of any of the above, or other memory.

(79) Volatile memory is memory whose state (and therefore the data stored on it) is indeterminate if power is interrupted to the device. Nonvolatile memory refers to memory whose state is determinate even if power is interrupted to the device. Dynamic volatile memory requires refreshing the data stored in the device to maintain state. One example of dynamic volatile memory includes DRAM (dynamic random access memory), or some variant such as synchronous DRAM (SDRAM). A memory subsystem as described herein may be compatible with a number of memory technologies, such as DDR3 (double data rate version 3, original release by JEDEC (Joint Electronic Device Engineering Council) on Jun. 27, 2007, currently on release 21), DDR4 (DDR version 4, JESD79-4 initial specification published in September 2012 by JEDEC), DDR4E (DDR version 4, extended, currently in discussion by JEDEC), LPDDR3 (low power DDR version 3, JESD209-3B, August 2013 by JEDEC), LPDDR4 (LOW POWER DOUBLE DATA RATE (LPDDR) version 4, JESD209-4, originally published by JEDEC in August 2014), WIO2 (Wide I/O 2 (WideIO2), JESD229-2, originally published by JEDEC in August 2014), HBM (HIGH BANDWIDTH MEMORY DRAM, JESD235, originally published by JEDEC in October 2013), DDR5 (DDR version 5, currently in discussion by JEDEC), LPDDR5, originally published by JEDEC in January 2020, HBM2 (HBM version 2), originally published by JEDEC in January 2020, or others or combinations of memory technologies, and technologies based on derivatives or extensions of such specifications. The JEDEC standards are available at www.jedec.org. The stripe mapping table **204** can be stored in volatile memory **726** or Non-Volatile Memory (NVM) **750**.

(80) Power source **740** provides power to the components of system **700**. More specifically, power source **740** typically interfaces to one or multiple power supplies **742** in system **700** to provide power to the components of system **700**. In one example, power supply **742** includes an AC to DC (alternating current to direct current) adapter to plug into a wall outlet. Such AC power can be renewable energy (e.g., solar power) power source **740**. In one example, power source **740** includes a DC power source, such as an external AC to DC converter. In one example, power source **740** or power supply **742** includes wireless charging hardware to charge via proximity to a charging field. In one example, power source **740** can include an internal battery or fuel cell source.

(81) Flow diagrams as illustrated herein provide examples of sequences of various process actions. The flow diagrams can indicate operations to be executed by a software or firmware routine, as well as physical operations. In one embodiment, a flow diagram can illustrate the state of a finite state machine (FSM), which can be implemented in hardware and/or software. Although shown in a particular sequence or order, unless otherwise specified, the order of the actions can be modified. Thus, the illustrated embodiments should be understood as an example, and the process can be performed in a different order, and some actions can be performed in parallel. Additionally, one or more actions can be omitted in various embodiments; thus, not all actions are required in every embodiment. Other process flows are possible.

(82) To the extent various operations or functions are described herein, they can be described or defined as software code, instructions, configuration, and/or data. The content can be directly executable (“object” or “executable” form), source code, or difference code (“delta” or “patch” code). The software content of the embodiments described herein can be provided via an article of

manufacture with the content stored thereon, or via a method of operating a communication interface to send data via the communication interface. A non-transitory machine-readable storage medium comprising a plurality of instructions stored thereon that, in response to being executed can cause a machine to perform the functions or operations described, and includes any mechanism that stores information in a form accessible by a machine (e.g., computing device, electronic system, etc.), such as recordable/non-recordable media (e.g., read only memory (ROM), random access memory (RAM), magnetic disk storage media, optical storage media, flash memory devices, etc.).

(83) A communication interface includes any mechanism that interfaces to any of a hardwired, wireless, optical, etc., medium to communicate to another device, such as a memory bus interface, a processor bus interface, an Internet connection, a disk controller, etc. The communication interface can be configured by providing configuration parameters and/or sending signals to prepare the communication interface to provide a data signal describing the software content. The communication interface can be accessed via one or more commands or signals sent to the communication interface.

(84) Various components described herein can be a means for performing the operations or functions described. Each component described herein includes software, hardware, or a combination of these. The components can be implemented as software modules, hardware modules, special-purpose hardware (e.g., application specific hardware, application specific integrated circuits (ASICs), digital signal processors (DSPs), etc.), embedded controllers, hardwired circuitry, etc.

(85) Besides what is described herein, various modifications can be made to the disclosed embodiments and implementations of the invention without departing from their scope.

(86) Therefore, the illustrations and examples herein should be construed in an illustrative, and not a restrictive sense. The scope of the invention should be measured solely by reference to the claims that follow.

Claims

1. An apparatus comprising: a plurality of solid state drives, each of the solid state drives including a non-volatile memory divided into a set of equally-sized zones; a Redundant Array of Independent Disks (RAID) controller to write a stripe including data strips and a parity strip to the plurality of solid state drives, each of the data strips and the parity strip stored in a different one of the solid state drives; and a stripe mapping table to include a stripe mapping entry for each stripe written to the plurality of solid state drives, the stripe mapping entry including numbers assigned to each of the solid state drives to store the data strips and the parity strip for the stripe.
2. The apparatus of claim 1, wherein each zone includes contiguous non-overlapping ranges of logical block addresses.
3. The apparatus of claim 1, wherein the RAID controller and plurality of solid state drives configured as a level 5 RAID system and each of the solid state drives is a member of the level 5 RAID system.
4. The apparatus of claim 3, wherein a number of solid state drives is 3, a first solid state drive to store a first data strip for the stripe, a second solid state drive to store a second data strip for the stripe and a third solid state drive to store the parity strip for the stripe, numbers assigned to the first solid state drive, second solid state drive and third solid state drive stored in the stripe mapping entry for the stripe in the stripe mapping table.
5. The apparatus of claim 3, wherein the stripe mapping table to include a stripe pointer to each stripe mapping entry, the stripe pointers reordered prior to a RAID rebuild to recover strips stored on a failed member drive in a correct order and write recovered strips sequentially to a replacement drive.

6. The apparatus of claim 1, wherein the RAID controller and plurality of solid state drives configured as a level 6 RAID system and each of the solid state drives is a member of the level 6 RAID system.
7. One or more non-transitory machine-readable storage medium comprising a plurality of instructions stored thereon that, in response to being executed, cause a system to: write, by a Redundant Array of Independent Disks (RAID) controller, a stripe including data strips and a parity strip to a plurality of solid state drives, each of the data strips and the parity strip stored in a different one of the solid state drives, each of the solid state drives including a non-volatile memory, the non-volatile memory divided into a set of equally-sized zones; and store, in a stripe mapping table, a stripe mapping entry for each stripe written to the plurality of solid state drives, the stripe mapping entry including numbers assigned to each of the solid state drives to store the data strips and the parity strip for the stripe.
8. The one or more non-transitory machine-readable storage medium of claim 7, wherein each zone includes contiguous non-overlapping ranges of logical block addresses.
9. The one or more non-transitory machine-readable storage medium of claim 7, wherein the RAID controller and plurality of solid state drives configured as a level 5 RAID system and each of the solid state drives is a member of the level 5 RAID system.
10. The one or more non-transitory machine-readable storage medium of claim 9, wherein a number of solid state drives is 3, a first solid state drive to store a first data strip for the stripe, a second solid state drive to store a second data strip for the stripe and a third solid state drive to store the parity strip for the stripe, numbers assigned to the first solid state drive, second solid state drive and third solid state drive stored in the stripe mapping entry for the stripe in the stripe mapping table.
11. The one or more non-transitory machine-readable storage medium of claim 9, wherein the stripe mapping table to include a stripe pointer to each stripe mapping entry, the stripe pointers reordered prior to a RAID rebuild to recover strips stored on a failed member drive in a correct order and write recovered strips sequentially to a replacement drive.
12. The one or more non-transitory machine-readable storage medium of claim 7, wherein the RAID controller and plurality of solid state drives configured as a level 6 RAID system and each of the solid state drives is a member of the level 6 RAID system.
13. A system comprising: a processor; a plurality of solid state drives, each of the solid state drives including a non-volatile memory divided into a set of equally-sized zones; a Redundant Array of Independent Disks (RAID) controller to write a stripe including data strips and a parity strip to the plurality of solid state drives, each of the data strips and the parity strip stored in a different one of the solid state drives; and a stripe mapping table to include a stripe mapping entry for each stripe written to the plurality of solid state drives, the stripe mapping entry including numbers assigned to each of the solid state drives to store the data strips and the parity strip for the stripe.
14. The system of claim 13, wherein each zone includes contiguous non-overlapping ranges of logical block addresses.
15. The system of claim 13, wherein the RAID controller and plurality of solid state drives configured as a level 5 RAID system and each of the solid state drives is a member of the level 5 RAID system.
16. The system of claim 15, wherein a number of solid state drives is 3, a first solid state drive to store a first data strip for the stripe, a second solid state drive to store a second data strip for the stripe and a third solid state drive to store the parity strip for the stripe, numbers assigned to the first solid state drive, second solid state drive and third solid state drive stored in the stripe mapping entry for the stripe in the stripe mapping table.
17. The system of claim 15, wherein the stripe mapping table to include a stripe pointer to each stripe mapping entry, the stripe pointers reordered prior to a RAID rebuild to recover strips stored on a failed member drive in a correct order and write recovered strips sequentially to a replacement

drive.

18. The system of claim 13, wherein the RAID controller and plurality of solid state drives configured as a level 6 RAID system and each of the solid state drives is a member of the level 6 RAID system.

19. The system of claim 13, further comprising: a power supply to provide power to the system.
